

Summary for 2024-10-22-Regression

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December 31, 2024

1 Summary

This lecture on Linear Programming explored the application of linear programming techniques to various regression and classification problems. The core concept is formulating seemingly non-linear problems as linear programs, enabling efficient solution using established linear programming algorithms.

* **Regression:** Regression methods, including least squares (L^2), least absolute deviations (L^1), and minimax (L^∞), were introduced, demonstrating how their solutions (mean, median, and mid-range, respectively) can be derived from minimizing different error metrics.

* **Linear Regression Formulation:** The lecture detailed how linear regression problems can be expressed using matrix notation, incorporating an error term to account for imperfect data fit.

* ** L^2 -Regression (Least Squares):** This method minimizes the sum of squared errors and can be solved using the normal equation, provided the relevant matrix is invertible.

* ** L^1 -Regression:** This method minimizes the sum of absolute errors and was shown to be reformulated as a linear program using slack variables.

* ** L^∞ -Regression:** This method minimizes the maximum absolute error and was also reformulated as a linear program using a slack variable to represent the maximum error.

* **Binary Classification:** The lecture extended the application of linear programming to binary classification problems, focusing on finding the optimal separating hyperplane.

* **Support Vector Machines (SVM):** The geometric interpretation of maximizing the minimum distance between classes was presented, leading to a linear programming formulation for finding the optimal separating hyperplane in an SVM context. This involved maximizing the margin subject to constraints on the signed distances of data points from the hyperplane. Normalization constraints were added to improve numerical stability.

* **Maximum Inscribed Circle Problem:** The lecture demonstrated how finding the maximum inscribed circle within a convex polyhedron can be formulated as a linear program. This involved expressing the distance from the circle's center to each constraint hyperplane and maximizing the circle's radius subject to these constraints. Redundant constraints were identified and removed for efficiency.

* **Smallest Ball Problem:** The lecture concluded by introducing the problem of finding the smallest ball containing a set of points in a d -dimensional space, highlighting its connection to convex programming.